DO NOW — RECALL

Lesson 138 · SQL SELECT, Filtering, Sorting and JOIN

Name: _____ Class: _____ Date: _____

1. What table is created to remove an EmployeeID→DeptID→DeptName transitive dependency?

2. When does a linear search stop?

3. Which is removed first from a FIFO queue?

4. What does ROM stand for?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Department(DeptID,DeptName), referenced by Employee.

2. Model answer

On a match or after examining all items.

3. Model answer

The longest-waiting item at the front.

4. Model answer

Read-only memory.

By the end, you can…

• SELECT chooses columns from a named relation.

• WHERE restricts rows; ORDER BY arranges results.

• Link keys using INNER JOIN ... ON.

Vocabulary — know these terms

SELECT — Returns specified columns/expressions.

WHERE — Filters rows according to a condition.

ORDER BY — Sorts returned rows by chosen column/expression.

JOIN — Combines rows from related tables using a join condition.

NEW LEARNING

Read, annotate and explain

Choose projection

SELECT chooses columns from a named relation.

Filter and order

WHERE restricts rows; ORDER BY arranges results.

Join related rows

Link keys using INNER JOIN ... ON.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

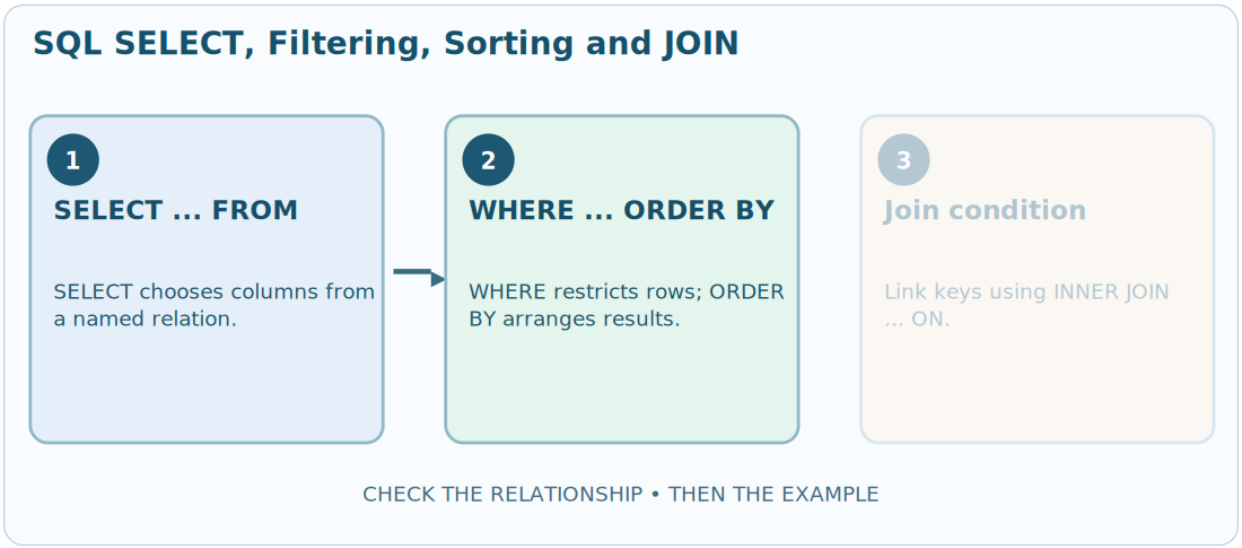
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SQL SELECT, Filtering, Sorting and JOIN; SELECT ... FROM then WHERE ... ORDER BY then Join condition

New Learning

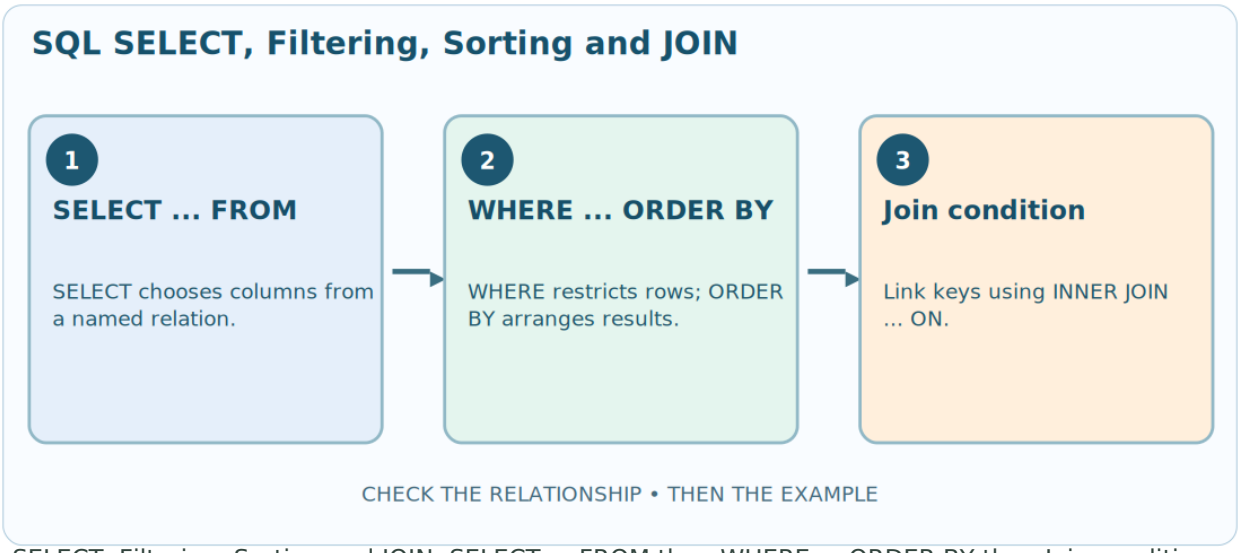


SQL SELECT, Filtering, Sorting and JOIN; SELECT ... FROM then WHERE ... ORDER BY then Join condition

New Learning · original lesson diagrams

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New Learning



SQL SELECT, Filtering, Sorting and JOIN; SELECT ... FROM then WHERE ... ORDER BY then Join condition

I DO — WORKED EXAMPLE

Follow the worked method

Orders(OrderID,CustomerID,Total) has totals 25,75,120. Write SQL listing orders over 50 with largest first.

Filter Total > 50

Find the given data and name the required result.

Select OrderID, Total

Explain why this intermediate stage is necessary.

Sort DESC 120,75

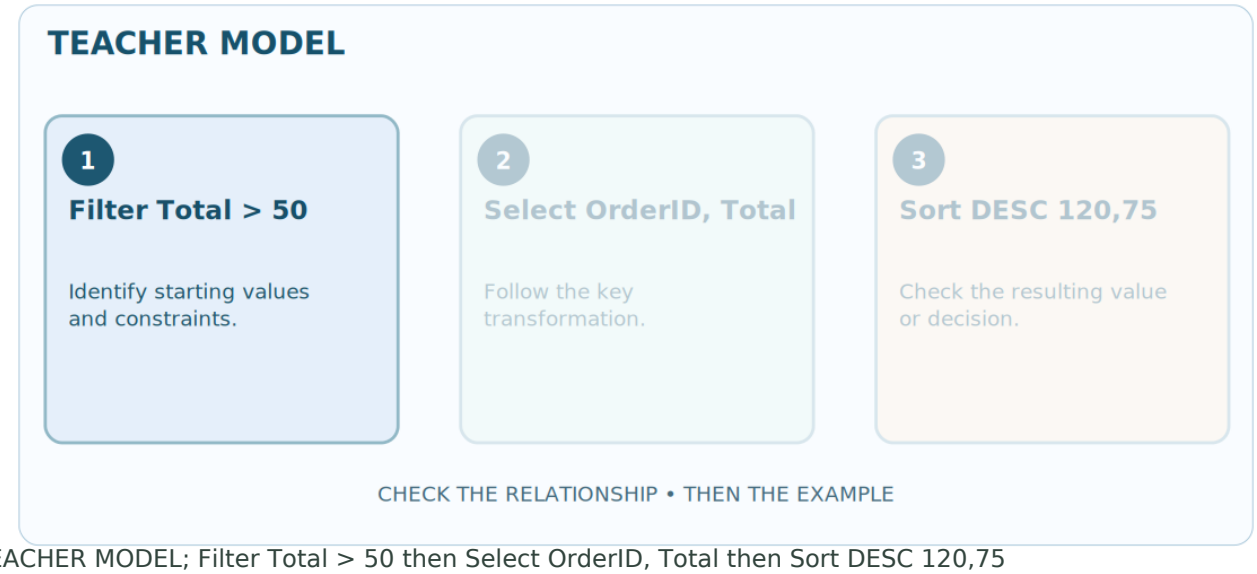
SELECT OrderID, Total FROM Orders WHERE Total > 50 ORDER BY Total DESC; returns orders with 120 then 75, excluding 25.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Orders(OrderID,CustomerID,Total) has totals 25,75,120. Write SQL listing orders over 50 with largest first.

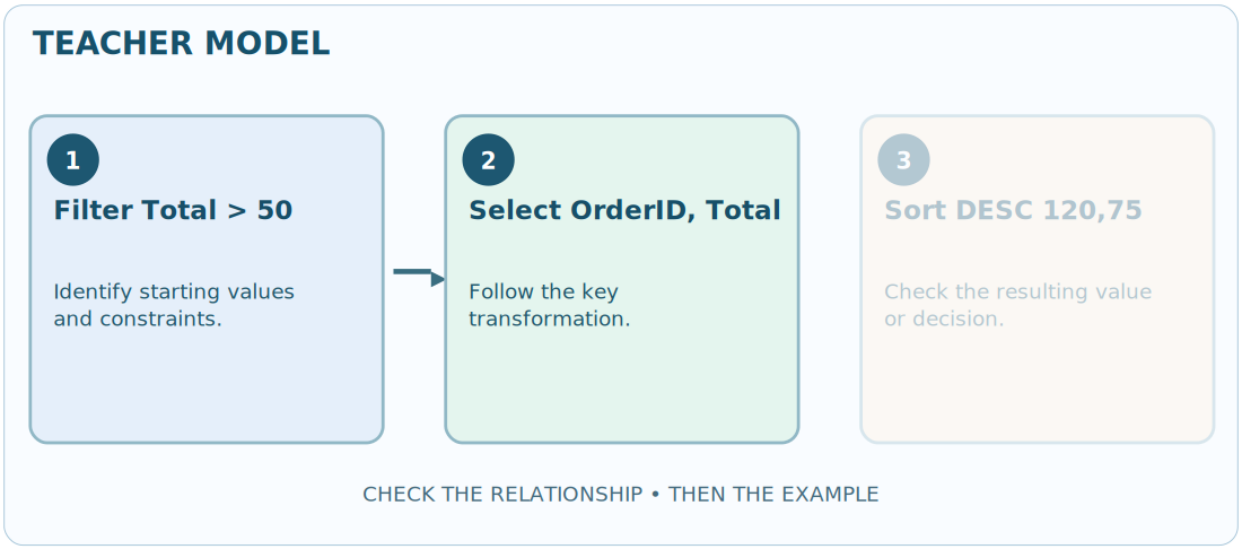
Find the given data and name the required result.



TEACHER MODEL; Filter Total > 50 then Select OrderID, Total then Sort DESC 120,75

Orders(OrderID,CustomerID,Total) has totals 25,75,120. Write SQL listing orders over 50 with largest first.

Find the given data and name the required result.



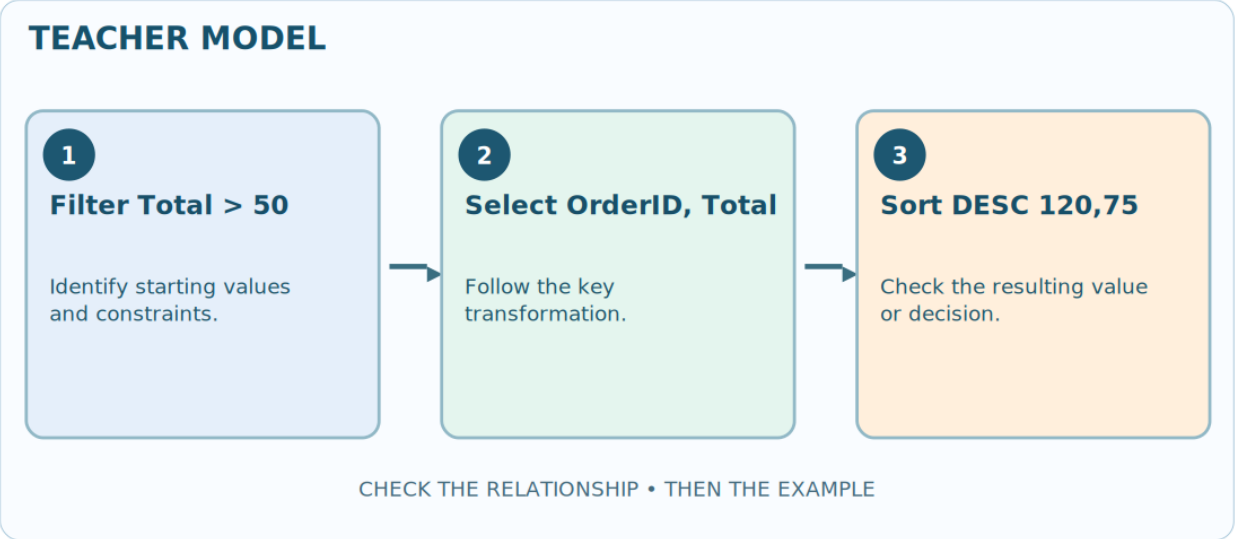
TEACHER MODEL; Filter Total > 50 then Select OrderID, Total then Sort DESC 120,75

I Do · original lesson diagrams

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TEACHER MODEL; Filter Total > 50 then Select OrderID, Total then Sort DESC 120,75

WE DO — GUIDED PRACTICE

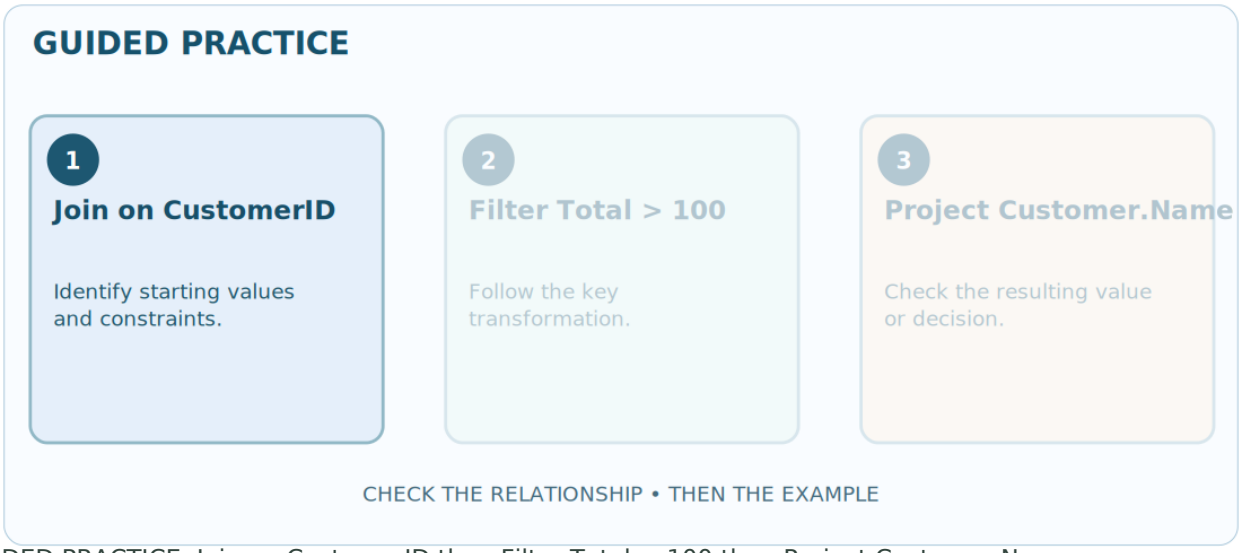
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Customer(CustomerID,Name), Orders(OrderID,CustomerID,Total). Show customer names for orders over 100.

Find the given data and name the required result.



GUIDED PRACTICE; Join on CustomerID then Filter Total > 100 then Project Customer.Name

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Join on CustomerID Find the given data and name the required result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write SQL listing Title and Price from Book where Price<10, sorted ascending by Price. [6]

Question 2

Explain the role of the ON clause in an INNER JOIN between Borrower and Loan. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write an INNER JOIN query listing Student.Name and Course.Title using Student, Enrolment and Course. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

ORDER BY always removes duplicate result rows.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

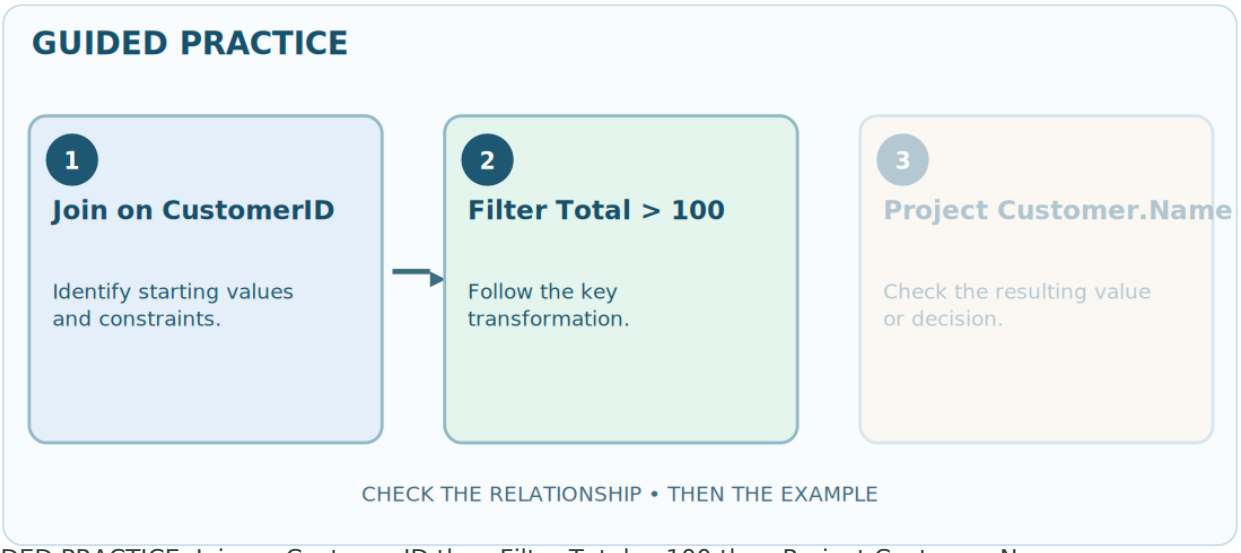
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Customer(CustomerID,Name), Orders(OrderID,CustomerID,Total). Show customer names for orders over 100.

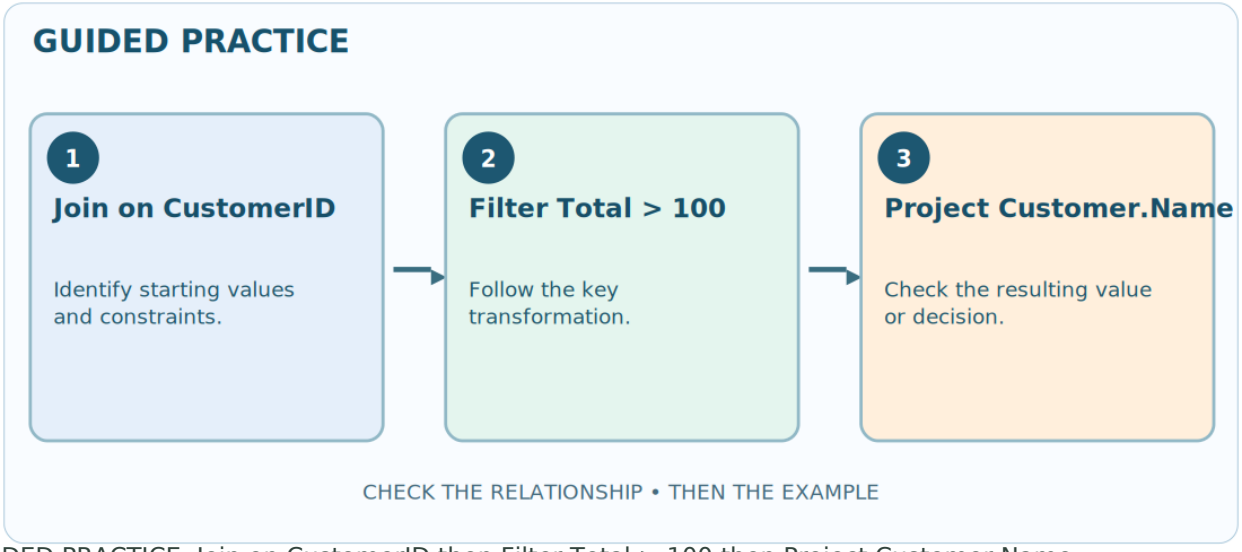
Find the given data and name the required result.



GUIDED PRACTICE; Join on CustomerID then Filter Total > 100 then Project Customer.Name

Customer(CustomerID,Name), Orders(OrderID,CustomerID,Total). Show customer names for orders over 100.

Find the given data and name the required result.



GUIDED PRACTICE; Join on CustomerID then Filter Total > 100 then Project Customer.Name

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Filter Total > 100

Explain why this intermediate stage is necessary.

Project Customer.Name

SELECT Customer.Name FROM Customer INNER JOIN Orders ON Customer.CustomerID = Orders.CustomerID WHERE Orders.Total > 100; Returns a row per matching order; DISTINCT can remove repeated customer names.

IMPROVE: Explain one correction to your own reasoning.